

Curriculum Vitae

Harim Chun

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EDUCATION

The Catholic University of Korea

Ph.D. Student, Department of Biomedical Sciences

Seoul, Korea

Mar. 2025–Present

Korea University College of Medicine

M.S., Department of Biomedical Sciences

GPA: 4.21/4.5 (96.7/100)

Seoul, Korea

Mar. 2021–Aug. 2024

University of Mississippi

Exchange Student, Biology

University, Mississippi, U.S.A.

Jan.–May 2019

Soongsil University

B.S., Systems Biomedical Science

Interdisciplinary Major in Big Data Computing | GPA: 4.11/4.5 (94.0/100)

Seoul, Korea

Mar. 2015–Feb. 2021

RESEARCH INTERESTS

Computational Cancer Genomics and Spatial Multi-Omics

Integrating spatial transcriptomics, single-cell multi-omics, and gene-regulatory-network analysis to investigate tumor-immune interactions and biomarkers of therapeutic response. Experienced in bulk RNA-seq, WES/WGS, scRNA-seq, scATAC-seq, Visium, and Xenium data analysis across collaborative cancer studies.

GPA AND ENGLISH SCORES

Graduate GPA (M.S.): 4.21/4.5; percentage equivalent: 96.7/100

Undergraduate GPA: 4.11/4.5; percentage equivalent: 94.0/100

TOEIC: 875/990

TEPS: 343/600

RESEARCH EXPERIENCE

PathFinderLab

The Catholic University of Korea College of Medicine and Korea University Anam Hospital

Graduate Student (Advisor: Sung Hak Lee, M.D., Ph.D.)

Research Associate

Seoul, Korea

Mar. 2025–Present

Sep. 2024–Feb. 2025

▪ Spatial Transcriptomics & Cancer Genomics

- Spatial architecture of neoadjuvant immunochemotherapy response in pleural mesothelioma
 - Integrating spatial transcriptomic and genomic data to characterize tumor-immune interactions and response-associated cellular states; presented findings at the AACR Annual Meeting 2026.

Choi Lab (Human Genetics and Genomics Lab)

Seoul, Korea

Korea University

Graduate Student (Advisor: Jungmin Choi, Ph.D.)

Mar. 2021–Aug. 2024

Undergraduate Research Intern (Advisor: Jungmin Choi, Ph.D.)

Jul. 2020–Feb. 2021

▪ Cancer Genomics

- Liver cancer pseudogene biomarker discovery
 - Analyzed 418 TCGA-LIHC samples, identified 465 differentially expressed pseudogenes, and prioritized 10 candidates associated with tumor grade and survival; co-first author of an IJMS article (2021).
- Metachronous gastric cancer biomarker discovery
 - Analyzed WES and bulk RNA-seq data from 46 patients, identified 13 differentially expressed genes, and prioritized KDF1, CDK1, and CREB5; co-first author in Cancer Medicine (2024).

▪ Single-Cell & Multi-Omics

- Transcriptional heterogeneity in the hematopoietic stem-cell niche
 - Analyzed droplet-based scRNA-seq data from Cxcr4 cKO and control mice and evaluated cell-state abundance and gene expression; contributed to Nature Communications (2022).
- Gene-regulatory networks associated with immune-checkpoint response
 - Integrated scRNA-seq and scATAC-seq data to identify response-associated regulatory networks in basal cell carcinoma.

▪ Bioinformatics

- Post-transcriptional regulation in *ctk1Δ* yeast
 - Identified 2,066 coding-sequence-level and 164 3'UTR-specific differentially expressed genes and characterized ribosomal and aging-related expression patterns.
- Mycobacterium tuberculosis WGS workflow
 - Built an automated alignment and GATK variant-calling workflow for drug-resistance inference and phylogenetic analysis.

Post-Genome Informatics Lab

Seoul, Korea

Research Intern, Soongsil University (Advisor: Sangsoo Kim, Ph.D.)

Aug. 2019–Jul. 2020

Insilicogen, Inc.

Yongin, Korea

Bioinformatics Intern

Jul.–Aug. 2019

- Built a Neo4j prototype for genomics-informed personalized nutrition and analyzed genomic and transcriptomic data using QIAGEN CLC Genomics Workbench.

RELEVANT SKILLS

Programming

R, Python

Computing

Unix/Linux, Hadoop, Apache Spark, GCP, AWS

TEACHING

KSBI-BIML 2024 , Korea Society for Bioinformatics Single-Cell Multi-Omics	Seoul, Korea Feb. 2024
KSBI-BIML 2023 , Korea Society for Bioinformatics Spatial Transcriptomics with R	Seoul, Korea Feb. 2023
The 19th KOGO Winter Workshop , Korea Genome Organization Spatial Transcriptomics with R	Seoul, Korea Feb. 2023
Training of Digital Healthcare Professionals , Korea University / INNOPOLIS NGS Data Analysis Pipeline — GATK Best-Practice Workflow	Seoul, Korea Oct. 2022
CoRE Tutoring , Soongsil University Hadoop File System and PySpark	Seoul, Korea Sep.–Dec. 2019

AWARDS AND HONORS

Asan Foundation Scholarship for Biomedical Science	Asan Foundation, 2026–2027
Excellence Poster Award	20th KOGO Winter Symposium, 2024
Brain Korea 21 (BK21) Scholarships	Korea Research Foundation, 2021–2022
Semester Honor Scholarship	Soongsil University, 2020
CoRE Tutoring Best Tutor Award	Soongsil University, 2020
Semester Honor Scholarship	Soongsil University, 2016

CONFERENCE PRESENTATIONS

Spatial architecture of neoadjuvant immunotherapy response in pleural mesothelioma	AACR 2026
Poster presentation	San Diego, USA, Apr. 2026
Rethinking molecular analysis of gastric cancer: insight into morphologic phenotype	AACR 2025
Poster presentation	Chicago, USA, Apr. 2025
Identification of molecular biomarkers for response to immune checkpoint inhibitors	KOGO Winter Symposium 2024
Poster presentation	Hongcheon, Korea, Feb. 2024 — Excellence Poster Award

PUBLICATIONS

Park JM, Na AY, Lee DY, Park S, Ahn SH, **Chun H**, Choi S, Park J, Lim C, Lee HS, Park HS, Choi J, Ryu HY. Ribosomal protein overexpression counteracts cellular senescence driven by dysfunctional RNA polymerase II CTD serine 2 phosphorylation. *Nature Communications*. Submitted (2025).

Lee H, Park Y, Jeon I, Kang D, **Chun H**, Choi J. GOREA: Unbiased functional enrichment interpretation. *Molecular Cells*. 2025.

Kim B*, **Chun H***, Lee J, Park M, Kwak Y, Kim JM, Kim SG, Ryu JK, Choi J, Cho SJ. Predictive biomarkers for metachronous gastric cancer development after endoscopic resection of early gastric cancer. *Cancer Medicine*. 2024.

Miao R, **Chun H**, Gomes AC, Choi J, Pereira J. Competition between hematopoietic stem and progenitor cells controls hematopoietic stem-cell compartment size. *Nature Communications*. 2022.

Lee J*, Hwang JH*, **Chun H***, Woo W, Oh S, Choi J, Kim LK. PLEKHA8P1 promotes tumor progression and indicates poor prognosis of liver cancer. *International Journal of Molecular Sciences*. 2021;22:7614.

* Co-first author